Unit 9, Lesson 11: Determining Surface Area Given Dimension Changes

Benchmark

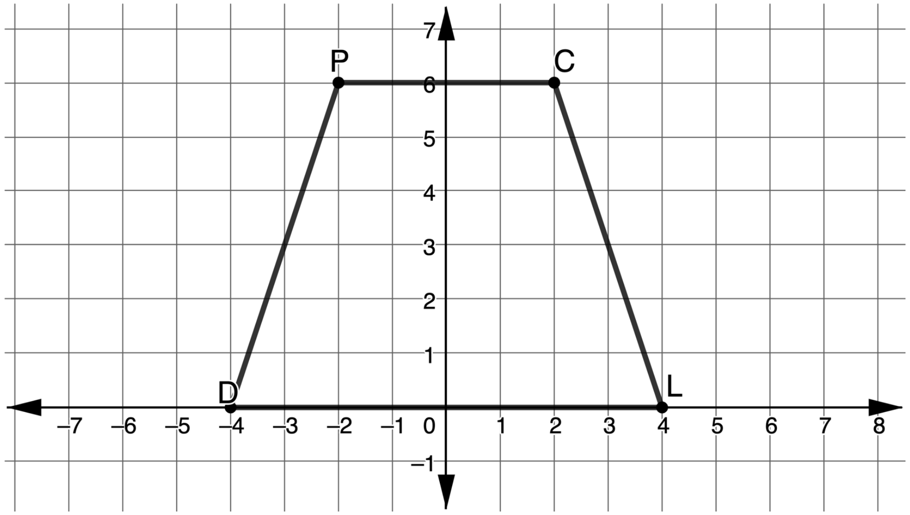
MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the area of two-dimensional figures and the surface area or volume of three-dimensional figures.

Learning Targets

- ☐ I can find the area of a figure when one or both dimensions are changed.
- ☐ I can find the surface area of a three-dimensional figure when one or both dimensions are changed.

9.11.1 Warm-Up: Bell Work

1. Trapezoid $DPCL$ is shown in the coordinate plane.



a. Find the area of $DPCL$.

b. Find the coordinates of the image of $MTRD$ given the scale factor in the table. Use the preimage coordinates to find the coordinates of the image. The dilation is centered at $(0,0)$.

Point	Preimage	Dilation $k = 6$ $(D'P'C'L')$
D		
P		
C		
L		

9.11.2 Exploration: Ryan's Renovations

Ryan is building a new house and wants to make the size of the kitchen larger. The current kitchen is 24 feet wide and 10 feet long. Ryan wants to make the new kitchen 20 feet long but leave the width the same.

1. Find the area of the original kitchen and the area of the new kitchen.

Area of the Original Kitchen	Area of the New Kitchen

2. How do the areas compare?

Ryan also wants to install a pool for his new house. The construction company gives Ryan three shape options for the new pool. Each pool's dimensions are given in feet (ft).

Pool 1	Pool 2	Pool 3
Cylindrical	Rectangular Prism	Cube
diameter $= 16$ ft height$= 12$ ft	length$= 30$ ft width$= 20$ ft height$= 12$ ft	Each side$= 15$ ft

3. Find the surface area of each of the shapes for the three pool options. Round to the nearest thousandth.

Pool 1	Pool 2	Pool 3

The construction company is giving Ryan more options for building a swimming pool. The three options are listed in the table.

Pool 4	Pool 5	Pool 6
Cylindrical	Rectangular Prism	Cube
diameter $= 8$ ft height $= 6$ ft	length $= 15$ ft width $= 10$ ft height $= 6$ ft	Each side $= 7.5$ ft

4. Describe how the dimensions of pair of pools are related.

Pair	Description
Pool 1 and Pool 4	
Pool 2 and Pool 5	
Pool 3 and Pool 6	

5. Find the surface area of each of the shapes for the three new options for a pool. Round to the nearest thousandth.

Pool 4	Pool 5	Pool 6

6. Record the surface area of all pool shapes in the table below.

Pool	Surface Area
1	
2	
3	
4	
5	
6	

The cost of building the swimming pool is dependent upon the amount of material. Since the top of each of the shapes would be water, determine the surface area of each pool that could be used to determine the cost.

7. Complete the table.

Pool	Surface Area for Material
1	
2	
3	
4	
5	
6	

8. If Ryan wants the least expensive pool, which one should he purchase?

9. Which pool would you purchase? State your reasoning.

9.11.3 Guided Practice: Ryan's House

Ryan is building a bathroom and needs to install either a bathtub or a shower. The options for each bathroom size are given.

Shower 1	Tub 1
length $= 4$ ft width $= 6$ ft height $= 12$ ft	length $= 4$ ft width $= 8$ ft height $= 10$ ft

1. In order to determine the amount of drywall, Ryan needs to find the surface area of each of the room options.

Shower 1	Tub 1

2. Ryan determines that the dimensions of the shower and tub are not up to code. The updated dimensions are shown. Use the new dimensions to determine the surface area of each room.

Shower 2	Tub 2
length $= 6$ ft width $= 9$ ft height $= 12$ f	length $= 6$ ft width $= 12$ ft height $= 15$ ft
Surface Area	**Surface Area**

Two of Shower 2's dimensions were changed by a factor of 1.5, whereas all of Tub 2's dimensions were changed by a factor of 1.5. Discuss with your partner on how to use the factor 1.5 and the number of changes in dimensions to find the new surface area from the original surface area.

3. Complete the statements.

________ $\times$ surface area of Shower 1 $=$ surface area of Shower 2

________ $\times$ surface area of Tub 1 $=$ surface area of Tub 2

9.11.4 Your Turn!

The living room of Ryan's house is 12 feet long and 18 feet wide.

1. The room is a rectangular shape. Find the area of Ryan's living room.

2. How would the area of the living room change if Ryan decided to enlarge the dimensions by 25%?

3. What is the scale factor for a dimension increase of 25%?

4. How did the area of the living room change with the new dimensions?

9.11.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.